Einstein repair with an integrable kernel: a sharp cyclic-cover theorem

24 September 2026. The positive-coercivity hypothesis in the uniformly local repair theorem excludes products of hyperbolic surfaces. Here all their genuine Teichmüller parameters are retained. On a specified cyclic-cover family we prove a complete nonlinear repair theorem with the correct quadratic loss, and actual near-Einstein metrics showing that loss cannot be replaced by a uniform constant. The reduced equation is proved to vanish; its solvability is not an assumption.

1. Objects, norms and statements

Fix closed oriented hyperbolic surfaces (B,b), (F,f), of curvature -1, and a primitive epimorphism $\pi_1(B)\to\mathbb{Z}$. Let (B_m,b_m) be the degree-m cyclic cover. Put

$$M_m = B_m\times F, \qquad h_m = b_m + f, \qquad \mathrm{Ric}_{h_m} = -h_m.$$

All constants below depend on these fixed surfaces, the cyclic cover and $0<\alpha<1$, but not on m. Use the uniform lifted product atlas to define $X_m = C^{3,\alpha}_{\mathrm{uloc}}$ and $Y_m = C^{1,\alpha}_{\mathrm{uloc}}$. These are supremum norms, not sums over charts. The choice of one extra derivative makes the global factor projections and gauge estimates explicit.

For a metric g, let

$$W_m(g)^k = g^{ij}\big(\Gamma(g)^k_{ij} - \Gamma(h_m)^k_{ij}\big), \quad \Phi_m(g) = \mathrm{Ric}_g + g - \tfrac{1}{2}\mathscr{L}_{W_m(g)}g. \tag{1}$$

The condition $W_m(g)=0$ says that the identity map to (M_m,h_m) is harmonic. At h_m,

$$D\Phi_m = L_m = \tfrac{1}{2}\Delta_{L,h_m} + I = \tfrac{1}{2}\nabla^*\nabla - \mathring{R}.$$

Write T_Σ for the transverse-traceless tensors on a hyperbolic surface, and set

$$K_m = T_{B_m}\oplus T_F = \ker L_m, \qquad \Pi_m = \text{the } L^2(h_m)\text{-projection onto } K_m. \tag{2}$$

Every vector in K_m is an actual infinitesimal factor deformation; its dimension grows with m. None is discarded.

Theorem A (quantified repair to the full Einstein moduli). There are $c,C,r_0>0$ and smooth families

$$H_m(k) = h_m + k + R_m(k), \qquad k\in K_m, \quad \|k\|_{X_m} < r_0, \tag{3}$$

of exact product Einstein metrics in harmonic gauge, with $\Pi_m R_m(k)=0$ and

$$\|R_m(k)\|_{X_m} \leqslant C\|k\|^2_{X_m}, \qquad \|DR_m(k)\| \leqslant C\|k\|_{X_m}. \tag{4}$$

If $W_m(g)=0$ and $\|g-h_m\|_{X_m} < cm^{-2}$, then

$$\boxed{\|g - H_m(\Pi_m(g-h_m))\|_{X_m} \leqslant Cm^2\|\mathrm{Ric}_g + g\|_{Y_m}.} \tag{5}$$

In particular every exact Einstein metric in this harmonic-gauge neighborhood belongs to (3). The assertion permits every tensor sector in the input metric, not only product or fiber-preserving perturbations.

Theorem B (the loss is realized nonlinearly). There are smooth metrics g_m, uniformly noncollapsed with uniformly bounded curvature derivatives of every fixed order, such that

$$\|g_m - h_m\|_{X_m} \leqslant Cm^{-2}, \qquad \|\operatorname{Ric}_{g_m} + g_m\|_{Y_m} \leqslant Cm^{-4}, \tag{6}$$

but, writing $\mathscr{E}(M_m) = \{h : \operatorname{Ric}_h = -h\}$,

$$\boxed{\inf_{h \in \mathscr{E}(M_m)} \|h - g_m\|_{X_m} \geqslant cm^{-2}.} \tag{7}$$

The infimum includes all normalized Einstein metrics and their diffeomorphic pullbacks, not just the displayed product family. Thus a uniform residual-to-distance estimate with exponent $\theta > 1/2$ is false on this family, even though all exact kernel directions are integrable. No optimality of the exponent 1/2 is asserted.

These manifolds already admit h_m. Theorem B is a limitation of uniform perturbative repair, not an Einstein-nonexistence result or a counterexample to Yau26.

2. The scalar inverse costs exactly m^2

Let $\Delta \geqslant 0$. We first prove

$$\|\Delta_{B_m}^{-1} v\|_{C^0} \leqslant Cm^2 \|v\|_{C^0}, \qquad \int_{B_m} v = 0, \tag{8}$$

with the inverse chosen to have mean zero. Cut the fixed base along a nonseparating curve dual to the epimorphism, and use bounded overlapping enlargements of its lifted cells. Adjacent cells and these enlargements have uniform Poincaré constants and bounded overlap. For a solution $\Delta u = v$, differences of consecutive cell means are bounded by the local L^2 gradient. Local elliptic estimates, after subtracting a cell mean, give

$$\sup_{\text{cell } j} |u - u_j| \leqslant C\Big(\|\nabla u\|_{L^2(\text{enlarged cell } j)} + \|v\|_\infty\Big).$$

To justify this estimate, cover the cell by a fixed chain of interior balls, apply the inhomogeneous local L^2-to-C^0 estimate on each ball, and use Poincaré on the enlargement. No estimate at an artificial cut boundary is used.

Summing along at most m cells and applying Cauchy–Schwarz yields

$$O := \operatorname{osc} u \leqslant C\sqrt{m}\|\nabla u\|_2 + C\|v\|_\infty.$$

Since $\operatorname{Vol} B_m = m \operatorname{Vol} B$, integration by parts, with any constant subtracted from u, gives

$$\|\nabla u\|_2^2 = \int uv \leqslant Cm\|v\|_\infty O.$$

Substitution and Young's inequality prove (8). Uniform local Schauder estimates then give its $C^{j,\alpha}$-to-$C^{j+2,\alpha}$ version, with the same m^2 loss.

For sharpness choose the harmonic one-form η in the primitive integral class. Its lift to the infinite cyclic cover is dz, with $z(\tau x) = z(x) + 1$ and $\Delta z = 0$. For $m \geqslant 2$,

$$v_m = \cos(2\pi z/m)$$

is a smooth function on B_m. Summing over the deck orbit proves $\int v_m = 0$, while $\|v_m\|_\infty = 1$. The chain rule gives

$$\Delta v_m = \frac{4\pi^2}{m^2}|\eta|^2 v_m, \qquad \|\Delta v_m\|_{C^{j,\alpha}_{\mathrm{uloc}}} \leqslant C_j m^{-2}. \tag{9}$$

All local derivatives of v_m are uniformly bounded. Thus the order in (8) is attained on an actual harmonic-circle construction, not merely inferred from a discrete graph model.

3. Uniform projections and the full transverse inverse

The classical product decomposition gives two slow channels

$$T_{B_m}\widehat{\otimes}L^2(F), \qquad L^2(B_m)\widehat{\otimes}T_F,$$

on which L_m is respectively $\frac{1}{2}\Delta_F$ and $\frac{1}{2}\Delta_{B_m}$. The remaining trace, conformal-Killing and mixed sectors have positive shifts. These are the complete sectors, as in Kröncke, Proposition 4.1 and Lemma 4.2 and the project's surface calculation, rescaled from curvature -3 to -1. In Kröncke's notation the Einstein operator $\Delta_E = \nabla^*\nabla - 2\mathring{R}$ is $2L_m$, not L_m. In our nonnegative Hodge convention, $\Delta_H = \nabla^*\nabla - 1$ on surface one-forms. Thus the exact intertwining needed below is

$$L_{B_m\times F}(D_B z) = D_B\Big(\tfrac{1}{2}\Delta_{H,B} + 1 + \tfrac{1}{2}\Delta_F\Big)z = D_B(A_B + \tfrac{1}{2}\Delta_F)z.$$

This also follows directly from the surface identities $L_B = D_B\delta_B$ on trace-free tensors and $\delta_B D_B = A_B$. The accompanying coordinate check verifies both identities for arbitrary one-form coefficients in the curvature--1 half-plane metric; no Hodge sign is inferred from a change of notation in the cited source.

We need Hölder, not just Hilbert-space, control of these projectors. For a hyperbolic surface let

$$D\xi = \delta^*\xi + \tfrac{1}{2}(\delta\xi)b, \qquad A = \delta D = \tfrac{1}{2}\nabla^*\nabla + \tfrac{1}{2}.$$

For a trace-free tensor s, its TT projection is $P_T s = s - DA^{-1}\delta s$. This formula has uniform bounds on the present surfaces, including as an operator from C^0 to local $C^{j,\alpha}$ for every fixed j.

Here is a proof which does not assume a bound for a large-rank projection. Put $w = A^{-1}\delta s$. The cyclic surfaces have $\operatorname{Vol} B_R(x) \leqslant C(1+R)$, by counting translates of a fixed cell in the infinite cyclic cover. Test $Aw = \delta s$ with $\phi_x^2 w$, where $\phi_x = e^{-a\,\mathrm{dist}(x,\cdot)}$ and $a > 0$ is fixed and sufficiently small. Integrate the divergence on the right onto $\phi_x^2 w$. The positive potential $1/2$, Young's inequality and $|d\phi_x| \leqslant a\phi_x$ give

$$\int \phi_x^2(|\nabla w|^2 + |w|^2) \leqslant C\int \phi_x^2|s|^2 \leqslant C\|s\|_\infty^2. \tag{10}$$

The last constant follows from the linear ball-growth bound. Thus $P_T s = s - Dw$ has uniformly bounded local L^2 norm. It is TT; the first-order operator $\delta : S_0^2 \to T^*\Sigma$ is elliptic in dimension two. Local homogeneous elliptic estimates therefore bound every derivative of $P_T s$ by this local L^2 norm. This proves the claim, first for smooth s, and then by density for the stated bounds. The fixed surface F has the same property.

The estimates persist with a parameter in the other factor: the projections commute with those derivatives, and their C^0-to-local- derivative bounds apply to parameter differences. Hence the factor projections, their product-sector extensions and Π_m are bounded on X_m, Y_m, uniformly in m. For Π_m, one additionally averages over the other factor; normalized averaging has norm at most one. This explains why a growing kernel dimension causes no projection constant here.

We claim that

$$\boxed{\|u\|_{X_m} \leqslant Cm^2\|L_m u\|_{Y_m}, \qquad \Pi_m u = 0.} \tag{11}$$

For the two slow channels, remove their true constant coefficient modes only, and apply (8) or the fixed scalar Green operator on F. This gives the required C^0 bound. For traces, the operator is $1 + \frac{1}{2}(\Delta_{B_m} + \Delta_F)$, so the maximum principle gives a uniform bound.

On a conformal-Killing sector write the source as $D_B w$, where $w = A_B^{-1}\delta_B s$. Its $C^{0,\alpha}$ norm on the product is bounded by $C\|s\|_{Y_m}$, using the positive-potential inverse and the preceding parameter estimates. Solve

$$(A_B + \tfrac{1}{2}\Delta_F)z = w, \qquad u = D_B z.$$

This connection operator has potential 1/2; the maximum principle and local Schauder estimates bound u in C^0 uniformly. The other conformal-Killing sector is identical with the factors interchanged. This retains its response rather than setting it to zero.

On the mixed bundle,

$$L_m = \tfrac{1}{2}\nabla^*\nabla, \qquad \langle L_m u, u\rangle_{L^2} \geqslant \|u\|_2^2,$$

because each factor's rough one-form Laplacian is $\Delta_H + 1$. Multiplication by a scalar preserves this entire bundle. The weighted identity in the preceding repair theorem thus applies with gap one and linear volume growth, giving a uniform local L^2 bound and then a C^0 bound. We do not apply weighted coercivity to the nonlocal kernel-orthogonal subspace: multiplication need not preserve that subspace.

Finally, the full local elliptic estimate

$$\|u\|_{X_m} \leqslant C(\|L_m u\|_{Y_m} + \|u\|_\infty)$$

proves (11) after summing the finitely many sector types. Fredholm theory gives surjectivity onto $K_m^\perp \cap Y_m$. Applying (9) to $u = v_m q$, for any fixed nonzero $q \in T_F$, shows that the operator norm in (11) is at least cm^2. These tensors are TT and orthogonal to the **entire** kernel (2).

4. Integration of the whole kernel, uniformly in the cover

We give the uniform version of the fixed-geometry argument in Round 18, §7. For $q \in T_{B_m}$ small in the uniform norm, the metric $b_m + q$ is positive definite and $K(b_m + q) + 1 = O(\|q\|^2)$, because the curvature variation of a TT tensor is zero. Solve

$$\Delta_{b_m+q}u + e^{2u} + K(b_m + q) = 0. \tag{12}$$

Its derivative at zero is $\Delta_{b_m} + 2$. The maximum principle and uniform local Schauder estimates give a uniform inverse; the nonlinear contraction therefore gives $u = O(\|q\|^2)$ on a uniform ball. Thus $e^{2u}(b_m+q)$ has curvature -1 and derivative q. TT elliptic estimates control all higher local derivatives of q by its uniform norm, so this construction and its first two parameter derivatives have uniform bounds. The same construction applies on F.

Place each hyperbolic family in harmonic-map gauge relative to its reference surface. The Jacobi operator at the identity is $\nabla^*\nabla + 1$. Its pointwise positive potential, the maximum principle and local Schauder theory give uniform inverse bounds. The implicit-function argument for maps to the fixed reference metric therefore has a uniform radius. Its first variation is zero along TT directions. Pushing the family forward by this small harmonic diffeomorphism preserves curvature and changes it only quadratically. Taking products gives actual Einstein metrics in the gauge (1).

Their kernel projection is $q+O(q^2)$, with uniformly bounded derivative. The inverse function theorem on K_m, using its uniform norm rather than coordinatewise parameter bounds, reparametrizes the family to give (3)–(4). Constants do not depend on its dimension.

Let $Q_m = I - \Pi_m$. The usual complementary equation is

$$Q_m \Phi_m(h_m + k + w) = 0, \qquad w \in K_m^{\perp}.$$

By (11) and the uniform quadratic Hölder estimate, it has a unique small solution in the radius cm^{-2} under consideration. The actual family already constructed is this graph: $w = R_m(k)$. Its remaining equation is therefore

$$\boxed{\Pi_m \Phi_m(h_m + k + R_m(k)) = 0}$$

identically. This proves compatibility of every reduced equation; it is stronger than omitting them or assuming integrability of only selected tangent vectors.

To prove (5), take $k = \Pi_m(g - h_m)$, $z = g - H_m(k) \in K_m^{\perp}$, and expand (1). The quadratic remainder N_m satisfies a uniform Lipschitz bound

$$\|N_m(a) - N_m(b)\|_{Y_m} \leqslant C(\|a\|_{X_m} + \|b\|_{X_m})\|a - b\|_{X_m}.$$

Since $\Phi_m(H_m(k)) = 0$, (11) gives

$$\|z\|_{X_m} \leqslant Cm^2\Big(\|Q_m \Phi_m(g)\|_{Y_m} + Ccm^{-2}\|z\|_{X_m}\Big).$$

Choose c small enough to absorb the last term. In harmonic gauge $\Phi_m(g) = \mathrm{Ric}_g + g$, proving Theorem A. The same argument classifies every nearby gauged zero, not merely the solutions obtained by a chosen iteration.

5. Actual small residuals separated from all Einstein outputs

Fix a nonzero $q \in T_F$, let $u_m = v_m q$ from (9), and first put $\tilde{g}_m(t) = h_m + tu_m$. The tensors u_m have uniformly bounded local derivatives of every fixed order, are TT, and satisfy

$$\Pi_m u_m = 0, \quad \|u_m\|_{X_m} \geqslant c_0 > 0, \quad \|L_m u_m\|_{Y_m} \leqslant Cm^{-2}.$$

Consequently

$$\|\mathrm{Ric}_{\tilde{g}_m(t)} + \tilde{g}_m(t)\|_{Y_m} \leqslant C(tm^{-2} + t^2). \tag{13}$$

This is the Taylor expansion of the actual Ricci tensor, not just a linearized forcing. Because $DW_m|_{h_m}(u_m) = 0$, the harmonic-map gauge adjustment described above is $O(t^2)$, with bounds in every fixed derivative norm. Denote the adjusted metric by $g_m(t)$. Then

$$W_m(g_m(t)) = 0, \qquad g_m(t) = h_m + tu_m + O_{X_m}(t^2), \tag{14}$$

and (13) still holds. Choose once and for all a sufficiently small $\theta_0 > 0$, and set $t = t_m = \theta_0 m^{-2}$. These metrics lie in Theorem A's neighborhood and satisfy (6).

For any family member $H_m(k)$ at distance $O(t_m)$ from $g_m(t_m)$, the bounded projection and (14) imply $\|k\|_{X_m} \leqslant Ct_m$. Equations (3)–(4) then give

$$Q_m(g_m(t_m) - H_m(k)) = t_m u_m + O_{X_m}(t_m^2).$$

Thus its norm is at least ct_m; boundedness of Q_m gives the same lower bound for the metric distance. Members farther away already obey that bound. By Theorem A this treats every gauged Einstein metric within its neighborhood.

For clarity, an arbitrary nearby Einstein output cannot evade this argument by a small change of gauge. If h is close to $g_m(t_m)$ in X_m, solve the harmonic map equation from (M_m, h) to (M_m, h_m) near the identity. At h_m its Jacobi operator on vector fields is $\nabla^*\nabla + 1$, so the inverse is uniform; small coefficient perturbations preserve that bound. Since the identity is already harmonic for $g_m(t_m)$, the solution F satisfies

$$\|F - \mathrm{id}\,\|_{C^{4,\alpha}_{\mathrm{uloc}}} \leqslant C\|h - g_m(t_m)\|_{X_m}.$$

It is a diffeomorphism. Its pushforward $\bar{h} = (F^{-1})^* h$ is Einstein and in the gauge (1). The metrics $g_m(t_m)$ have uniformly bounded $C^{4,\alpha}$ norms, so writing

$$\bar{h} - g_m = (F^{-1})^*(h - g_m) + \left((F^{-1})^* g_m - g_m\right)$$

gives $\|\bar{h} - g_m\|_{X_m} \leqslant C\|h - g_m\|_{X_m}$, with no derivative loss in this comparison. Taking θ_0 smaller if necessary puts $\bar{h}$ inside the classified neighborhood whenever the proposed distance is less than a fixed small multiple of t_m. The gauged lower bound contradicts such a distance. Outputs outside that small ball trivially satisfy (7). This proves Theorem B.

Combining (6)–(7), for every $\theta > 1/2$,

$$\frac{\mathrm{dist}_{X_m}(g_m, \mathscr{E}(M_m))}{\|\mathrm{Ric}_{g_m} + g_m\|^{\theta}_{Y_m}} \longrightarrow \infty.$$

In particular the m^2 loss in the linear residual estimate (5) has matching nonlinear examples.

6. Mathematical position

The product spectrum and the integration of surface Teichmüller directions are classical; see Kröncke above and Rupflin–Topping on horizontal hyperbolic metrics. Round 18 already proved local completeness at each fixed finite-monodromy geometry. The increment here is the **uniformly quantified cyclic regime**: a sharp Hölder-space transverse inverse, an explicit nonlinear radius, all reduced equations solved by the actual full moduli, and a nonlinear obstruction to volume-independent strong repair estimates.

There is no contradiction with the preceding positive-coercivity theorem. This family has exact kernels and scalar slow channels, although its local geometry is uniformly bounded and its ball growth is only linear. Nor does this exclude a different existence proof, a weaker repair modulus, or a theorem retaining additional global control of the slow coefficients. It identifies a real geometric quantity that such a theorem must handle; it does not identify another absent gauge zero mode.

`check_integrable_kernel_cyclic_repair.py` checks the curvature-−1 surface differential identities, the harmonic-circle chain rule, deck-orbit cancellation and the nonlinear scaling. It also checks the negative control that localization need not preserve kernel orthogonality. These finite calculations do not verify the cellwise elliptic estimates, uniform moduli integration or nonlinear existence argument.